

# John Feczko

412-328-1626 | jpfeczko@yahoo.com | www.linkedin.com/in/john-feczko | 319 Chestnut Street, Sewickley, PA 15143

## Education

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**Villanova University – College of Engineering**, Villanova, PA

2027

*Bachelor of Science in Mechanical Engineering, Minor in AI, Concentration in Dynamic Systems, Minor in Humanities*

GPA: 3.94      Dean's List, all semesters

**Senior Research; AI Robot Guide**

Fall 2026

- This project will study how AI acting through a mobile robotic system can interpret environmental information and generate motion decisions using a combination of sensors, embedded computing hardware, and AI-based software tools
- Research tools and techniques will include the use of an NVIDIA DGX Spark, Jetson Orin boxes, APIs, and Optical Character Recognition (OCR), along with GPS and camera data

**Mechanical Engineering Sophomore Research Scholar; Smart Textiles**

2024

- Researched nanoparticle assembly and atomic monolayer analysis, including the enhanced detection of PFAS particles and fireproofing of ordinary textiles through nanoparticle coating

**Class Projects; Hands-on design projects within coursework, involving CAD, programming, prototyping, and fabrication, including:**

- Built a functioning web app using generative AI orchestration, user GPS data, and RAG documentation to make informed responses on the history of the user's location
- Designed a cost-effective automatic fire suppressant system intended for arid-climate refugee tent camps
- Conducted full lifecycle development of a small robotic "Beetle-Bot" system from concept to fabrication, electrical integration, testing, and post-failure analysis

## Work Experience

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**Arconic Corporation; Manufacturing Engineering Internship**

Summer 2026

- Work on projects for the senior process engineer at an aluminum processing plant to improve plant operating efficiency
- Investigated wastewater treatment system and identified design improvements estimated to save nearly \$1M per year
- Analyzed the current natural gas infrastructure and provided recommendations on optimal meter positions

**Handshake AI Fellowship; AI Model Trainer**

2025 – Present

- Conduct AI feedback research as an independent contractor with industry AI development companies
- Develop and evaluate domain-specific prompts to assess the performance of large language models (LLMs)
- Analyze LLM outputs for scientific accuracy, clarity, and depth in specialized subfields

**BCI Engineering; Summer Intern, Leetsdale, PA**

2023 – 2024

- Designed parts, drawings, assemblies, Bill of Materials (BOM) and applied General Drafting & Tolerancing (GD&T) in CAD
- Conducted rapid prototyping for new tools and models to improve quality assurance efficiency
- Created Quality Assurance reviews using First Article Inspection (FAI) reports and load-test data summary reports

## Activities & Leadership

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Baja SAE Offroad Vehicle Steer Team Lead

2026 – 2027

American Society of Mechanical Engineers (ASME)

2023 – Present

Physics II Tutor

2025 – Present

Faith & Reason Scholar Program

2024 – Present

Engineering service trip to Kenya

2025

Villanova Men's Rowing

2023 – 2025

## Technical Skills

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- Claude Code, prompt engineering, Retrieval Augmented Generation (RAG), hosting AI agents, MCP tools, model finetuning
- SolidWorks, including simple Finite Element Analysis (FEA) in mechanical stress and heat transfer simulations
- Autodesk Inventor, including the creation of parts, assemblies, drawings, and BOMs
- AutoCAD, Fusion, FreeCAD, MATLAB, Maple, Python, Arduino, and C++
- Advanced with standard and resin 3D printing, laser cutting/engraving, and traditional machining (lathe, mill, saw, etc.)
- Experience with CNC milling, CNC lathing, and WASER waterjet cutting